PART 4 Instructions

Now, we explain how your payoff will be determined.

The bottom line is that **the more accurate your estimate is, the more likely you will earn a higher payoff.**

How is your payoff determined?

Step 1: At the end of Part 2, your penalty score for each of your estimates will be calculated in the following way. You will receive a higher penalty score from Box A as your estimate concering yourself deviates more from the actual number of Green balls you received. You will receive a higher penalty score from Box B as your estimate concerning your teammate deviates more from the actual number of Green balls your teammate received.

$$Penalty\ Score\ from\ Box\ A = \sum_{i=1}^{10} \left(tokens\ in\ bin\ i - 100 * I_i^A\right)^2$$

$$Penalty\ Score\ from\ Box\ B = \sum_{i=1}^{10} \left(tokens\ in\ bin\ i - 100 * I_i^B\right)^2$$

- $i = 1, \ldots, 10$ refers to the different bins in each of Box A and Box B.
- I_i^A equals 1 for the bin in Box A which contains the actual number of Green balls that you received, and 0 otherwise.
- I_i^B equals 1 for the bin in Box B which contains the actual number of Green balls that your teammate received, and 0 otherwise.

Step 2: Then, a number X^A is drawn randomly from the interval [0,20000]. Any number within the interval [0,20000] is equally likely to be selected. Independent of X^A, a number X^B is also drawn in the same way. If your Penalty Score from Box A is less than or equal to the random number X^A, you will earn 30 Experimental Dollars. Otherwise, you will earn zero. If your Penalty Score from Box B is less than or equal to the random number X^B, you will earn 30 Experimental Dollars. Otherwise, you will earn zero. In other words, the higher penalty score you get, the lower probability it is that you earn 30 Experimental Dollars for each of your estimates.

To help your understanding, suppose you received 55 Green balls in Part 1. Instead of informing you of the number of Green balls you received in the experiment, we ask you to estimate how many Green balls you received by allocating 100 tokens to each of 10 bins in Box A. Notice that, in this example, the bin that contains the actual number of Green balls you received is *bin "51-60"*.

Consider the following scenarios.

- **Scenario 1:**
  You feel very confident that you received more than (or equal to) 51 but less than (or equal to) 60 Green balls in Part 1. Therefore, you allocated all the 100 tokens to the “51-60” bin. If this is the case, you receive no

penalty because your estimate is very accurate. You earn 30 Experimental Dollars with certainty (100 percent chance).

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	0	100	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(100 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*0*

- **Scenario 2:**

  You feel very confident that you received more than (or equal to) 41 but less than (or equal to) 50 Green balls in Part 1. Therefore, you allocated all the 100 tokens to the "41-50" bin. If this is the case, your penalty score is 20000 (the highest possible penalty score) because your estimate is completely wrong. You have a zero percent chance of earning 30 Experimental Dollars.

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	100	0	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(100 - 0)^2$	$(0 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*20000*

- **Scenario 3:**

  You guess that you received 41-50 Green balls with a 50 percent chance, and 51-60 Green balls with a 50 percent chance. Therefore, you allocated 50 tokens to the "41-50" bin and the rest of 50 tokens to the "51-60" bin. If this is the case, your penalty score is 5000. You earn 30 Experimental Dollars with a 75 percent chance.

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	50	50	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(50 - 0)^2$	$(50 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*5000*

- **Scenario 4:**

  You guess that you received 41-50 Green balls with a 30 percent chance, and 51-60 Green balls with a 70 percent chance. Therefore, you allocated 30 tokens to the "41-50" bin and the rest of 70 tokens to the "51-60" bin. If this is the case, your penalty score is 1800. You earn 30 Experimental Dollars with a 91 percent chance.

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	30	70	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(30 - 0)^2$	$(70 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*1800*

- **Scenario 5:**

  You decided to just randomly distribute the 100 tokens to each bin. If this is the case, your penalty score is 9000. You earn 30 Experimental Dollars with a 55 percent chance.

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	10	10	10	10	10	10	10	10	10	10	
	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 100)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	*9000*

In summary, you will earn 30 Experimental Dollars with certainty if your estimate is very accurate. There will be no chance to earn 30 Experimental Dollars if your estimate is completely wrong. **To increase the chance of earning 30 Experimental Dollars, you should provide an estimate as accurate as possible.**

The same rule applies to your estimate concerning your teammate. **To increase the chance of earning 30 Experimental Dollars, you should provide an estimate as accurate as possible.**

You can review the instructions by clicking the "Back" button.

Once you get familiar with Part 4 instructions, click the "Next" button.

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